

Example EDA Report - sales_and_customers

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This report is generated by EDA LLM Assistant: exploratory analysis for tabular data, with explicit notes on which methods are used and how they are applied.

Report output (how to read this document)

- Input data source (from config): sales_and_customers.csv
- Sections follow: dictionary -> column intelligence -> structure -> validation -> transformations -> visualizations -> supervised (if target set) -> other notes -> provenance.
- Plots are embedded as relative paths under the output folder.
- Method notes appear as blockquotes (lines starting with >) under each subsection.

1. Data dictionary

We build a data dictionary table for every column: dtype, distinct count, missing rate, an inferred meaning from name patterns (token heuristics, not domain truth), and whether the column is treated as a candidate independent variable or the configured target. You can override any column with dictionary: in the YAML config (column name -> definition).

column	dtype	n_unique	missing_pct	in_analysis	meaning	role_in_eda
CustomerID	int64	200	0	no	Inferred: likely Identifier (surrogate or business key) (from column name pattern).	excluded from analysis (identifier / manual)
Age	int64	55	0	yes	Inferred: likely Age or aging-related measure (from column name pattern).	numeric (candidate independent)
Income	int64	199	0	yes	Inferred: likely Monetary amount or income (from column name pattern).	numeric (candidate independent)
Annual_Spend	float64	200	0	yes	Inferred: likely Monetary amount or income (from column name pattern).	numeric (candidate independent)

column	dtype	n_unique	missing_pct	in_analysis	meaning	role_in_eda
Number_of_Visits	int64	29	0	yes	Inferred: likely Count or numeric quantity (from column name pattern).	numeric (candidate independent)
Preferred_Channel	object	3	0	yes	Inferred: likely Channel or source category (from column name pattern).	categorical (candidate independent)
Customer_Satisfaction	object	3	0	yes	Inferred: likely Score or satisfaction level (from column name pattern).	dependent (target)

If inferred meanings are wrong, replace them in config under dictionary: or in your organization's official data dictionary.

2. Column intelligence

Column intelligence (heuristic). We label each loaded column with a semantic guess (money-like, count-like, high-cardinality text, etc.) using dtype, name tokens, and cardinality. Separately, on string columns we scan a sample for PII-like patterns (email, phone, SSN-shaped strings, Luhn-passing digit strings that may resemble payment cards). These are screening hints, not legal classification-expect false positives/negatives.

column	in_analysis	semantic_guess	semantic_note	pii_risk_flags	pii_scan_note	n_unique	cardinality_ratio
CustomerID	no	surrogate_or_manual_excluded	Column excluded from analysis (often an ID).	none		200	1
Age	yes	continuous_or_count_numeric	Generic numeric; inspect domain.	none		55	0.275

column	in_analysis	semantic_guess	semantic_note	pii_risk_flags	pii_scan_note	n_unique	cardinality_ratio
Income	yes	monetary_like	Name suggests money; verify currency/units.	none		199	0.995
Annual_Spend	yes	monetary_like	Name suggests money; verify currency/units.	none		200	1
Number_of_Visits	yes	count_like_integer	Integer with count-like name.	none		29	0.145
Preferred_Channel	yes	nominal_or_low_cardinality_categorical	Typical categorical / string labels.	none		3	0.015
Customer_Satisfaction	yes	nominal_or_low_cardinality_categorical	Typical categorical / string labels.	none		3	0.015

- Sample export: column_intelligence.csv

3. Data structure

Method - excluding identifiers from analysis. Surrogate keys (e.g. CustomerID, id, order_id) are usually not modeling features: they inflate numeric summaries, distort correlations/outliers, and leak train/test identity if misused. By default we drop columns that match ID-like name patterns before validation plots, correlations, IQR outliers, and distributional plots. They still appear in the data dictionary with in_analysis = no. The configured columns.target is never auto-dropped. Disable with columns.auto_exclude_id_columns: false or add explicit drops with exclude_from_analysis.

Excluded from statistical analysis (identifiers / manual): CustomerID

3.1 Data summary

- Rows (n): 200
- Columns (p): 6
- Approx. memory: 0.03 MB (pandas memory_usage(deep=True))
- Sample export: sample_rows.csv

3.2 Numeric vs categorical (and other roles)

Method - numeric vs categorical vs datetime. We rely on pandas dtypes after load: select_dtypes for numeric (int, float), boolean, datetime64, and object/category strings. Columns listed under columns.datetime_columns in the config are coerced with pandas.to_datetime(..., errors='coerce'), which parses valid dates and turns failures into missing (NaT).

Role	Columns (count)	Names
Numeric	4	Age, Annual_Spend, Income, Number_of_Visits
Categorical (object/category)	2	Customer_Satisfaction, Preferred_Channel
Boolean	0	-
Datetime	0	-

3.3 Dependent vs independent variables

Dependent vs independent variables. The config marks Customer_Satisfaction as the dependent (target) variable for supervised analysis. All other analyzed columns are treated as candidate independent variables (predictors / features). Causal interpretation still requires study design, not EDA alone.

4. Data validation

4.1 Missing values

*Method - missing values. We count missing entries with pandas.isna() on each column (treats NaN/None and nullable types as missing). For each column we report missing_count and missing_pct = missing_count / n_rows * 100. Bar and heatmap visuals help scan which columns and rows are affected.*

No missing values detected.

4.2 Duplicate records

Method - duplicate rows. We use DataFrame.duplicated() with default settings (all columns, keep first occurrence as non-duplicate). The count is the number of rows that repeat an earlier row exactly.

- Duplicate rows: 0
- Duplicate %: 0.0000%

4.3 Anomalies / consistency (heuristic checks)

Method - basic error / consistency checks (heuristic). We do not know your domain rules, so we flag:

- Constant columns (?1 distinct non-null value) - often IDs wrongly typed or placeholders.
- High missing rate - columns above a configurable fraction of missing cells.
- Mixed-type object columns - sample of values converts to numeric for some but not all; may indicate dirty strings or combined formats.

These are starting points for manual or business validation, not automatic corrections.

No heuristic flags from the built-in checks (see methodology for limits).

4.4 Outliers (IQR fences)

Method - outliers (Tukey / IQR rule). For each numeric column we compute:

- $Q1 = 25\text{th percentile}$, $Q3 = 75\text{th percentile}$.
- $IQR = Q3 - Q1$ (interquartile range: spread of the middle 50% of the data).
- Lower fence = $Q1 - 1.5 \times IQR$, upper fence = $Q3 + 1.5 \times IQR$.

Any row where the value is below the lower fence or above the upper fence is counted as an IQR-based outlier for that column. If IQR is 0 (or undefined), we skip flagging for that column to avoid degenerate fences. This rule is descriptive, not a claim that those points are data entry errors.

column	outlier_count	outlier_pct
Age	0	0
Annual_Spend	0	0
Income	0	0
Number_of_Visits	0	0

5. Data transformation

Transformations. This report separates: (1) applied in this run - only what the pipeline actually changed (e.g. datetime coercion); (2) suggested - common next steps from skew, missingness, or cardinality (you choose whether to implement in modeling pipelines).

5.1 Applied in this run

- Identifier / manual exclusion: dropped from analysis (still in dictionary): CustomerID.

5.2 Suggested next steps (not applied automatically)

No generic transformation suggestions from heuristics, or dataset is small/simple.

6. Data visualization

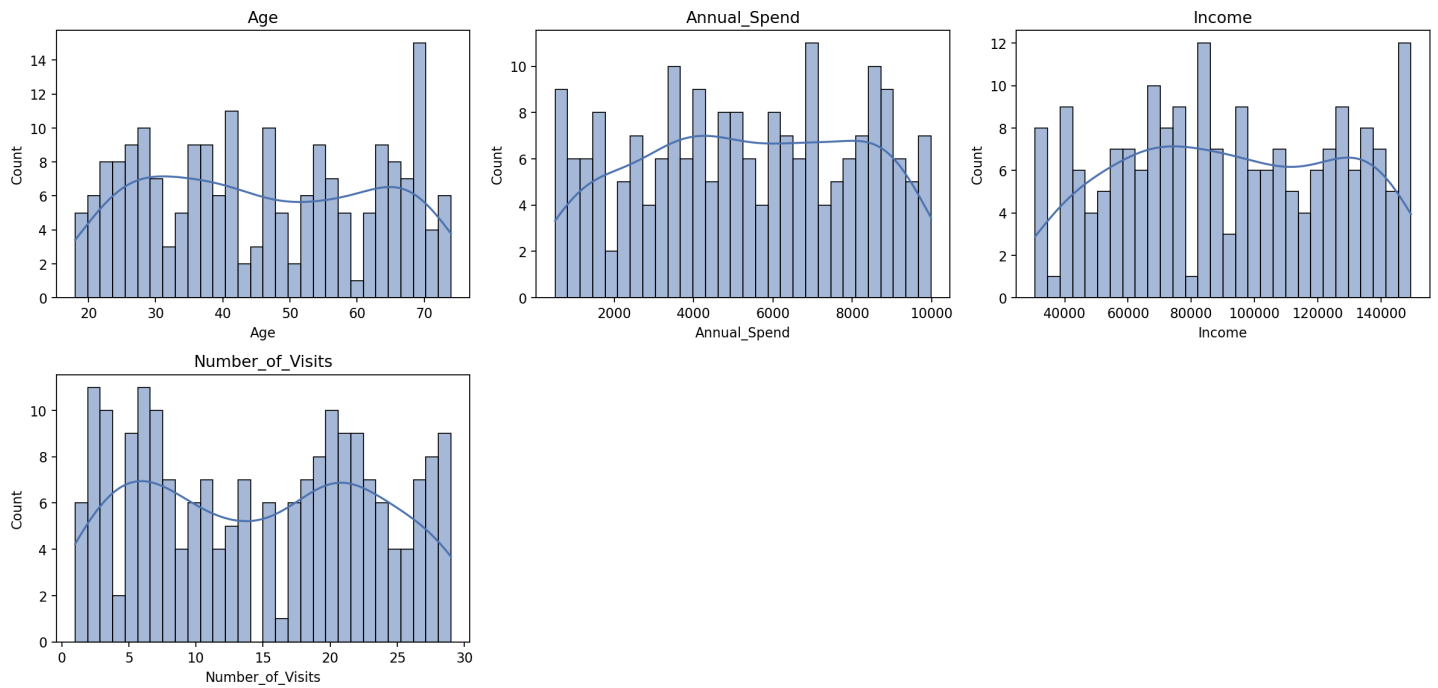
6.1 Numeric distributions

Method - numeric summaries. For numeric columns we use `DataFrame.describe()` with extra percentiles (e.g. 1%, 5%, 50%, 95%, 99%). That gives count, mean, std, min/max, and distribution quantiles. Skewness and kurtosis use the usual pandas `.skew()` and `.kurtosis()` on numeric columns.

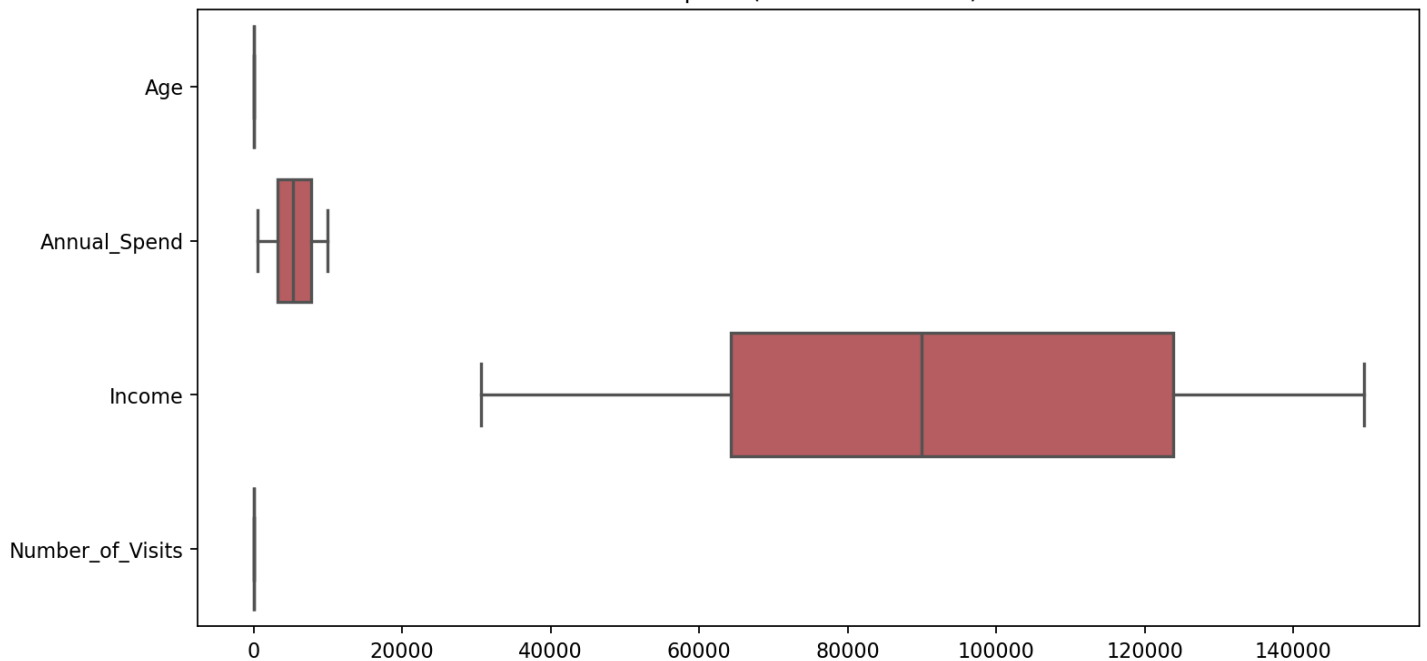
Plots - numeric. Histograms with KDE (`seaborn.histplot, kde=True`) show shape and density. Horizontal boxplots summarize quartiles and show points beyond the whiskers (useful alongside the IQR counts above).

column	mean	std	min	p50	max
Age	45.855	16.7324	18	45	74
Annual_Spend	5338.69	2743.75	504.2	5269.09	9984.84
Income	92201.7	34532.1	30591	90003.5	149489
Number_of_Visits	14.715	8.71571	1	15	29

Numeric distributions (hist + KDE)



Boxplots (numeric columns)



6.2 Categorical distributions

Method - categorical top values. For object/category columns we compute value_counts (including missing as its own bucket) and show the top N categories by frequency.

Plots - categorical. Bar charts show the top categories per column (same logic as value_counts, limited to a max number of columns for readability).

Customer_Satisfaction

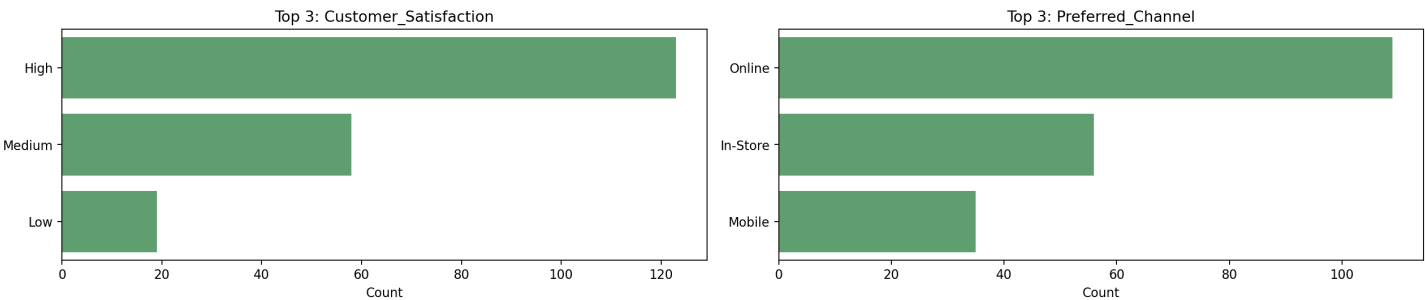
value	count
High	123
Medium	58

value	count
Low	19

Preferred_Channel

value	count
Online	109
In-Store	56
Mobile	35

Categorical top values



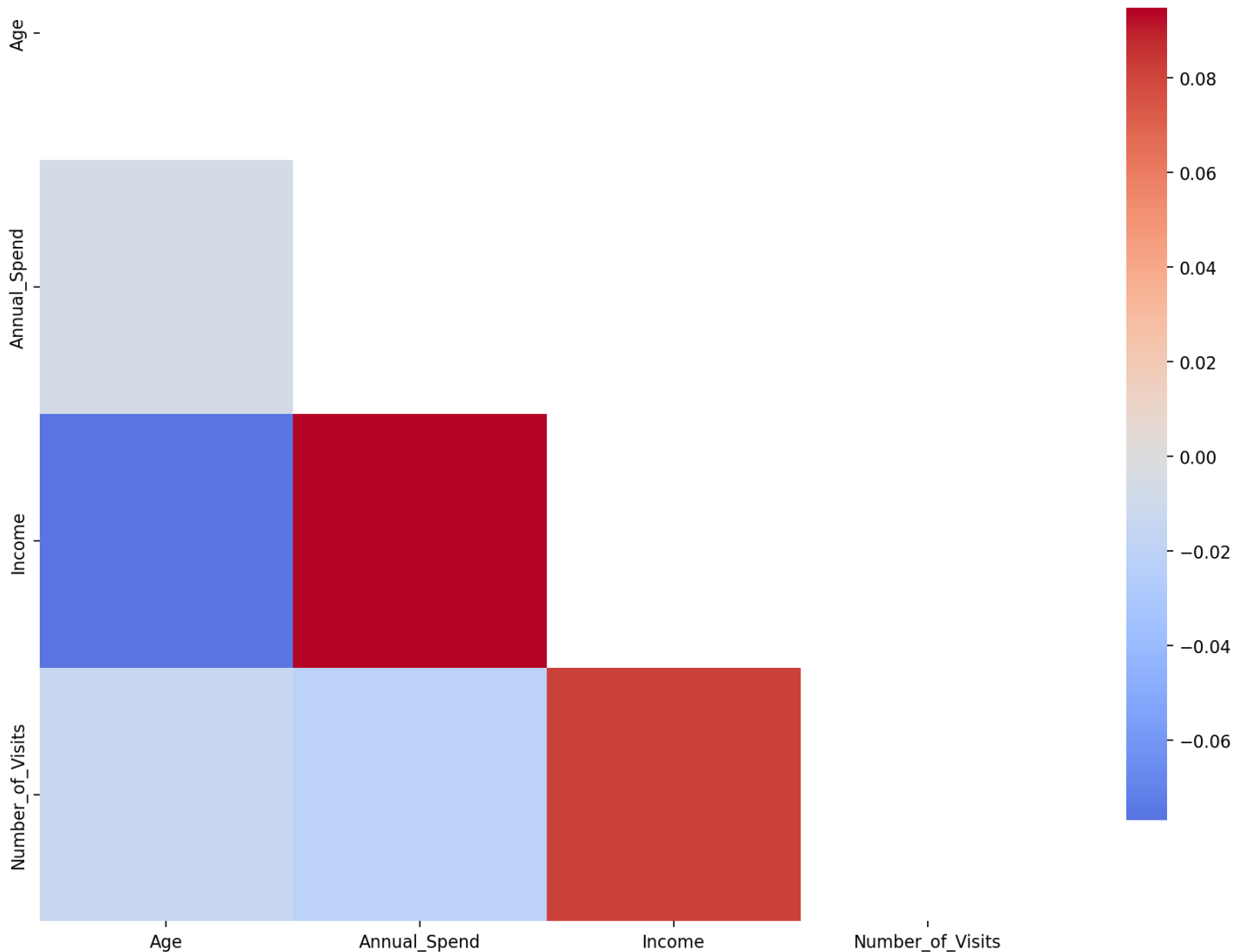
6.3 Correlations

Method - Pearson correlation. For numeric columns we compute the pairwise Pearson correlation matrix with `DataFrame.corr()` (default method `pearson`). Each entry is between -1 and 1: linear association, not causation. We flag pairs with absolute correlation $\geq$ `report.corr_threshold` as "high" for review.

Plots - correlation. The heatmap shows the Pearson matrix; the upper triangle is masked so each pair appears once and the diagonal is easier to scan.

No pairs with $|r| \geq$ threshold (edit `report.corr_threshold`; current logic uses absolute Pearson r).

Correlation heatmap (upper triangle masked)



7. Supervised / target-driven EDA

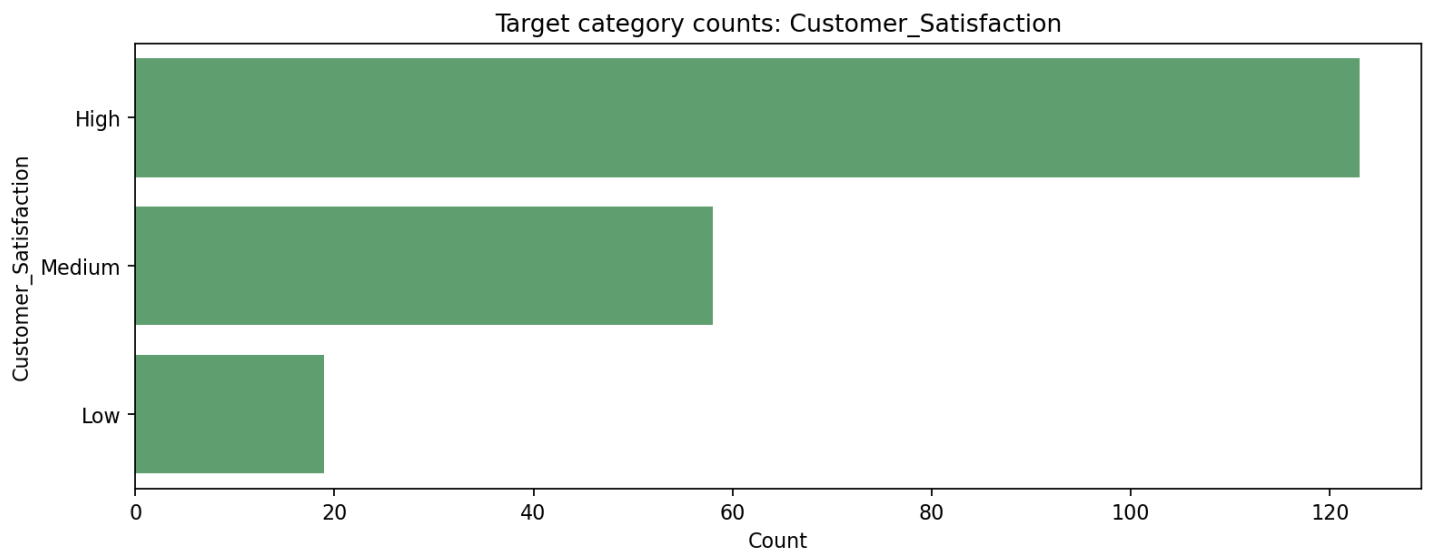
Supervised / target-driven EDA. With columns.target set, we add tables that relate the target to other features: for a numeric target, group summaries and Kruskal-Wallis tests vs categoricals (nonparametric, independent groups); for a categorical target, numeric summaries by target level and chi-square tests vs other categoricals (independence on contingency tables). Assumptions are stated with each test. High-cardinality categoricals are skipped for chi-square (configurable cap) to avoid unstable tables.

Target column: Customer_Satisfaction

Type (for this section): categorical

7.1 Target profile

Customer_Satisfaction	count
High	123
Medium	58
Low	19



Numeric feature Age by target level

Customer_Satisfaction	count	mean	median	std
Low	19	53.5263	55	13.8499
High	123	45.5935	43	17.1066
Medium	58	43.8966	40.5	16.336

Numeric feature Annual_Spend by target level

Customer_Satisfaction	count	mean	median	std
High	123	5462.29	5585.44	2673.28
Medium	58	5325.66	5086.35	2780.7
Low	19	4578.32	4209.55	3096.79

Numeric feature Income by target level

Customer_Satisfaction	count	mean	median	std
High	123	95488.5	95392	35543.9
Low	19	94261.9	94234	32118.2
Medium	58	84556.6	77442.5	32390.5

Numeric feature Number_of_Visits by target level

Customer_Satisfaction	count	mean	median	std
Low	19	17.8947	22	8.66599
Medium	58	16.5172	19	9.44837
High	123	13.374	12	8.13553

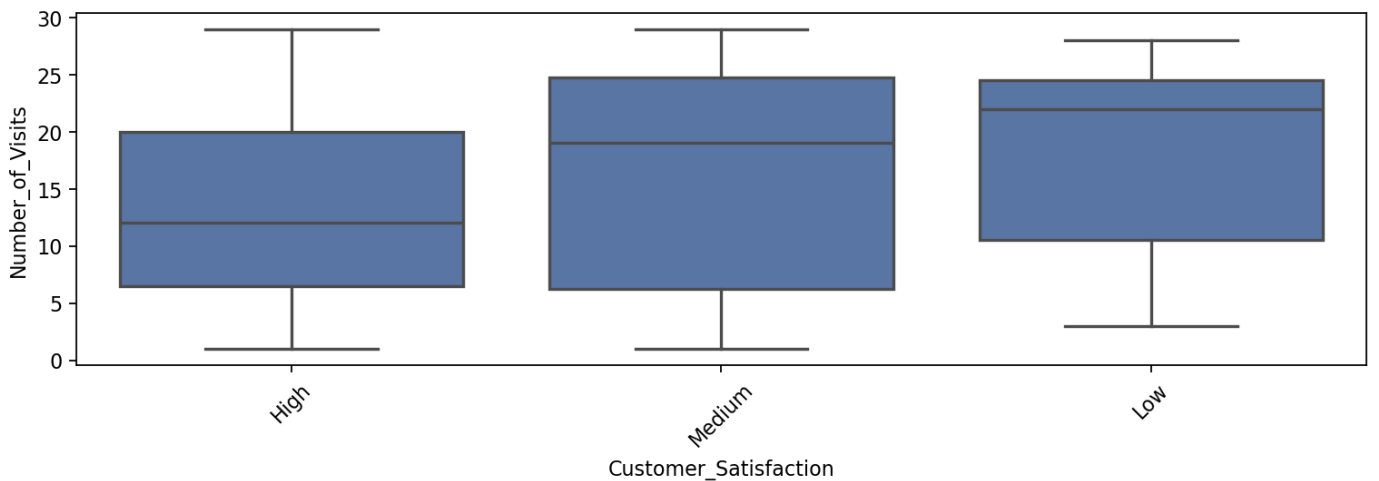
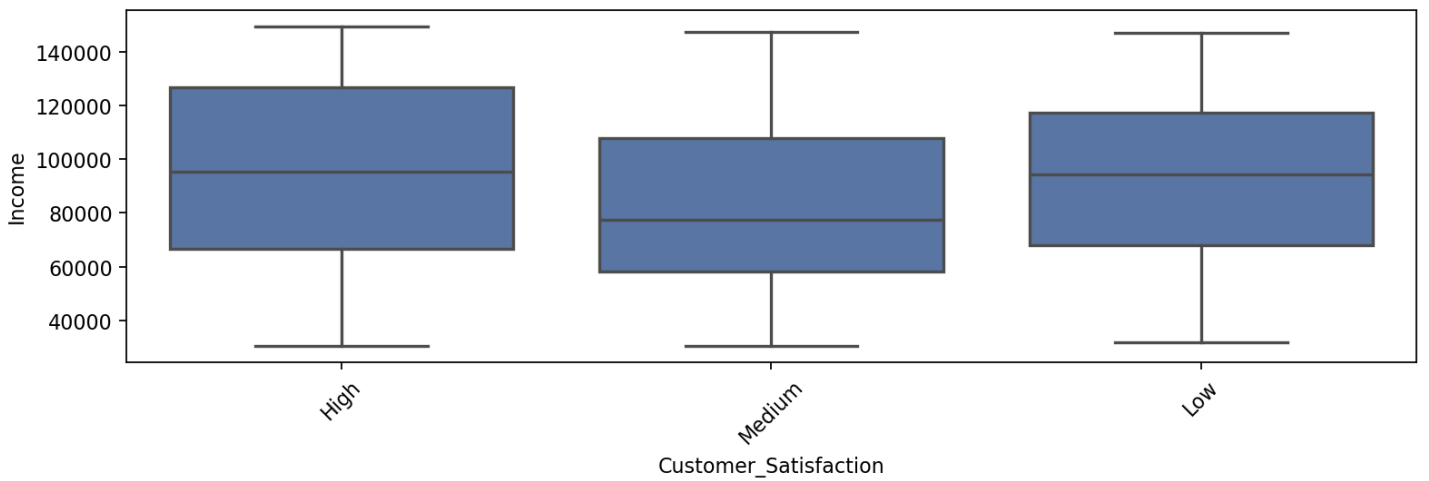
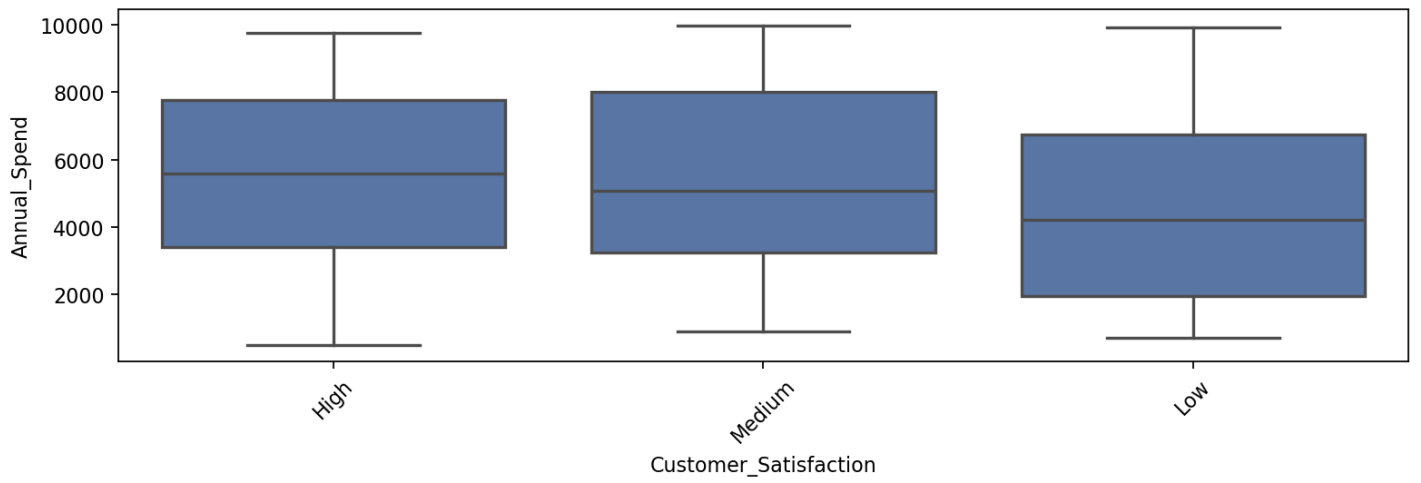
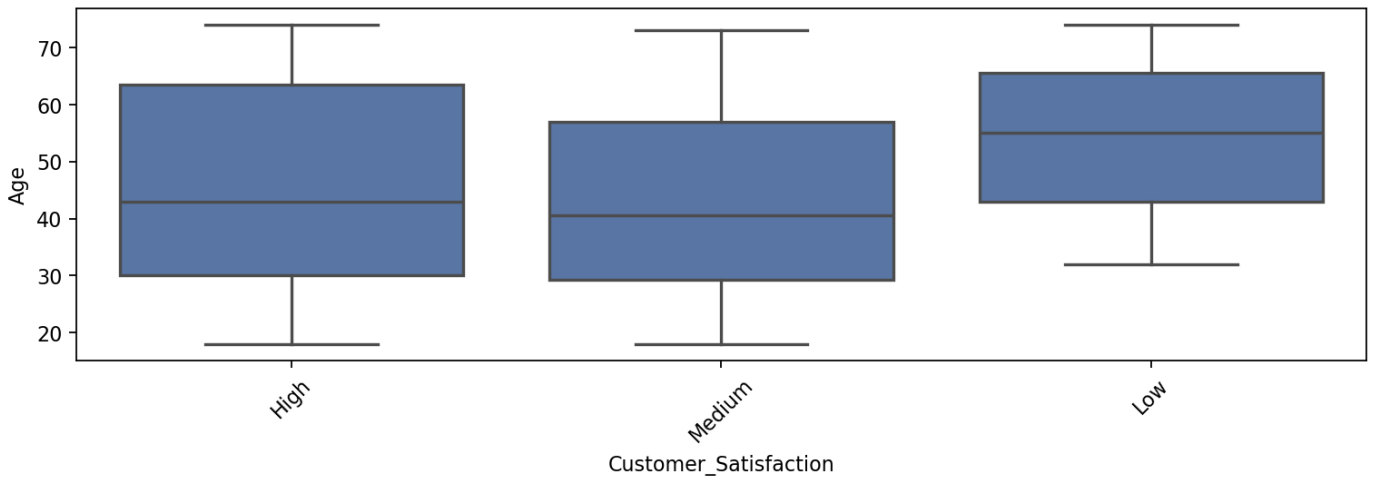
Chi-square tests (target vs categorical features)

Preferred_Channel

- chi-squared = 3.0983, p = 0.5415, dof = 4, table shape 3x3, 11.1% of cells have expected count < 5.

Pearson chi-square test of independence on contingency table. Assumes independent rows; expected counts should be >5 in most cells for asymptotic p-values to be reliable. High cardinality inflates chi² and power.

Numeric features by categorical target `Customer\_Satisfaction`



8. Other considerations

Other considerations. ? Sampling: if report.sample_rows is set, plots use a random sample for speed; tables and validation use the full dataframe unless noted. ? Correlation: Pearson assumes roughly linear relationships; for skewed or ordinal data consider Spearman or dedicated tests. ? Causality: association in EDA does not imply cause. ? Privacy: reports may contain sample rows; scrub sensitive columns before sharing.

LLM-generated narrative (optional)

Generated from a compact JSON summary of EDA tables, not from raw row-level data.

LLM summary enabled, but env var OPENAI_API_KEY is not set. Skipping LLM summary.

9. Output provenance

Key	Value
generated_at_utc	2026-03-26 16:35:38 UTC
python	3.11.5
pandas	2.1.1
numpy	1.24.3
matplotlib	3.8.0
seaborn	0.12.2
scipy	1.11.3
eda_llm_assistant	0.1.0